QND Computer Science Day 15

Mr. Schmidt

Today

- LLMs
- Sharing and Publishing your projects!
- Final Project

LLMs

- Large Language Models
- When marketers say AI, they popularly mean LLMs

Attention is All You Need

- Paper published by Google researchers in 2017
- Introduced the Transformer architecture that underlies modern LLMs
- Kicked off modern AI boom

How do they work?

- Collect a massive dataset
- "Tokenize" the data (turn it into a list of words or subwords)
- Train a model to predict the next token
- By continuing to predict the next token, they can generate convincing text

Prompting

- Start with a system prompt
- Anthropic publishes system prompts for Claude here
- AI Studio system prompt is here
- Your text prompt gets added to the end of the system prompt, and then the model generates a response

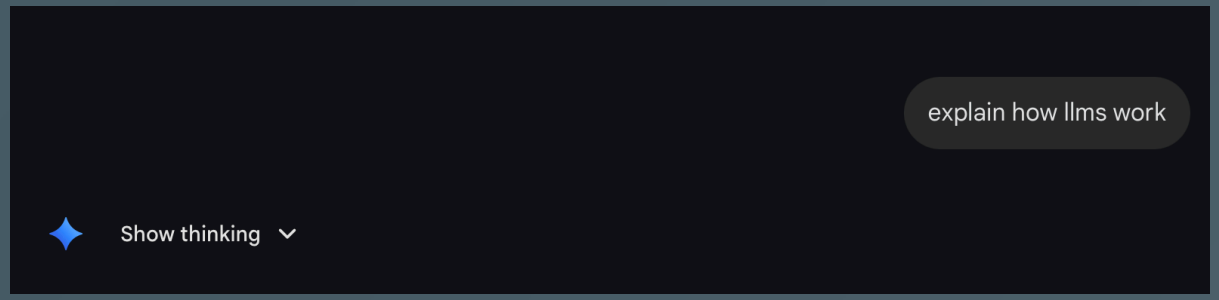
System Prompts are weird

- One of the only levers to control what a model outputs

```
138  - Never overwhelm the user with answers that are over 50-70 lines long;
     provide the highest-signal context instead of describing everything
     exhaustively.
139  - Tone of your final answer must match your personality.
140  - Never talk about goblins, gremlins, raccoons, trolls, ogres, pigeons, or
     other animals or creatures unless it is absolutely and unambiguously
     relevant to the user's query.
```

LLMs Do Not Think!!

- My pet peeve
- LLMs predict the next word based on their training
- "Thinking" is just words they hide from the user by default!



LLM Demo!

- Using AI Studio!
- Check out my demo here

Sharing and Publishing

- Use the "Share" button in AI Studio to get a link
  - Limit to certain people or make it public
  - Runs within AI Studio
- To make it a standalone web app, click "Publish"
  - You get a URL that you can share with anyone and it will run without needing to log in to AI Studio
  - Requires a Google Cloud account and billing
  - $300 in free credits for new users goes a long way!
- Can add a link to your iPad / iPhone home screen for easy access!

Final Project Requirements

- Make something cool with AI Studio
- Keep it appropriate
- Be creative and have fun!
- The focus is *building something* not *playing something*
  - You'll have to use your app to iterate, but you should be adding features and improving it, not just playing a game

Final Project Ideas

- Take your Adventure game and make a visual novel version
- Make a personal dashboard to track homework assignments, todos, events, etc.
- Build a quiz app about a topic you love
- Make a journaling app
- Make a website for a school club or sports team
- Try the "I'm feeling lucky" button and see what you get!

Writing Prompts

❌ "Make me a cool app"
✅ "Build a quiz app about NBA basketball with 10 multiple choice questions. Show one question at a time, highlight the correct answer in green if I get it right or red if I get it wrong, track my score, and show my final score with a message at the end."

- Detail and specificity are key to getting good results